PIZZA SALES ANALYSIS

A MySQL Query Portfolio · 13 Business Questions Solved

BASIC • INTERMEDIATE • ADVANCED

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Dataset: Pizza Place Sales | Tool: MySQL Workbench

01

Basic Queries

Foundational aggregations to understand the scale of the business

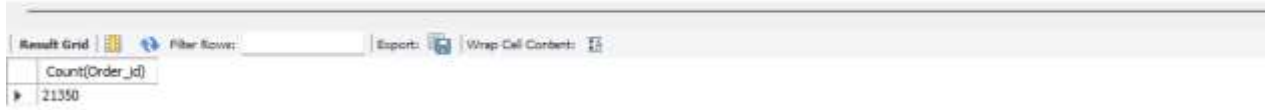
BASIC · Q1

Retrieve the total number of orders placed.

KEY INSIGHT

The dataset covers 21,350 orders, giving a solid, high-volume base for reliable trend and revenue analysis across the full order history.

```
1  -- Retrieve the total number of orders placed.
2
3· Select Count(Order_id) From orders;
```



BASIC · Q2

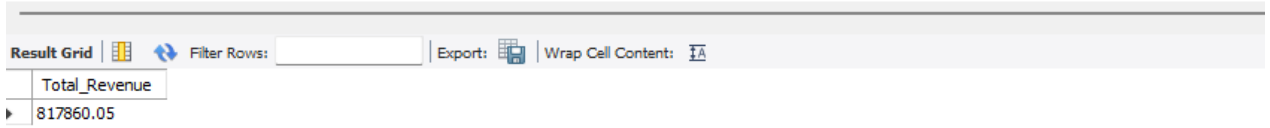
Calculate the total revenue generated from pizza sales.

KEY INSIGHT

Total revenue across all orders comes to $817,860.05, calculated by joining order line items with pizza prices and summing quantity x price.

```sql
-- Calculate the total revenue generated from pizza sales.

SELECT
    ROUND(SUM(orders_details.QUANTITY * pizzas.price),
            2) AS Total_Revenue
FROM
    orders_details
        JOIN
    pizzas ON orders_details.PIZZA_ID = pizzas.pizza_id;
```



BASIC · Q3

Identify the highest-priced pizza.

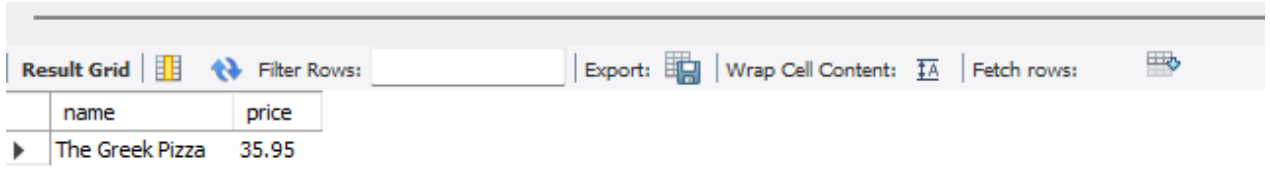
KEY INSIGHT

The Greek Pizza is the most expensive item on the menu at $35.95, making it a natural candidate for premium-tier promotions.

```sql
-- Identify the highest-priced pizza.

SELECT
    py.name, p.price
FROM
    pizza_types py
        JOIN
    pizzas p ON py.pizza_type_id = p.pizza_type_id
ORDER BY p.price DESC
LIMIT 1;

```



name	price
The Greek Pizza	35.95

BASIC · Q4

Identify the most common pizza size ordered.

KEY INSIGHT

Large (L) is the clear customer favorite with 18,956 units ordered, ahead of Medium (15,635) and Small (14,403). XL and XXL sizes are niche.

```sql
-- Identify the most common pizza size ordered.

SELECT
    pizzas.size,
    Sum(orders_details.QUANTITY) AS order_count
FROM
    pizzas
        JOIN
    orders_details ON pizzas.pizza_id = orders_details.PIZZA_ID
GROUP BY pizzas.size
ORDER BY order_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

size	order_count
L	18956
M	15635
S	14403
XL	552
XXL	28

BASIC · Q5

List the top 5 most ordered pizza types along with their quantities.

KEY INSIGHT

The Classic Deluxe Pizza leads with 2,453 orders, closely trailed by Barbecue Chicken, Hawaiian, Pepperoni, and Thai Chicken — all within ~80 units of each other.

```sql
-- List the top 5 most ordered pizza types along with their quantities.
SELECT
    pizza_types.name,
    SUM(orders_details.QUANTITY) AS order_count
FROM
    orders_details
        JOIN
    pizzas ON orders_details.PIZZA_ID = pizzas.pizza_id
        JOIN
    pizza_types ON pizzas.pizza_type_id = pizza_types.pizza_type_id
GROUP BY pizza_types.name
ORDER BY order_count DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

name	order_count
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

02

Intermediate Queries

Multi-table joins and grouped analysis to reveal category and time patterns

INTERMEDIATE · Q6

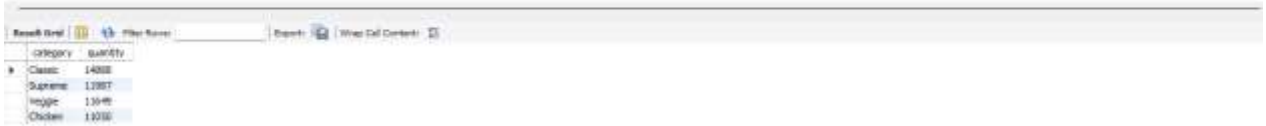
Join the necessary tables to find the total quantity of each pizza category ordered.

KEY INSIGHT

Classic pizzas lead category volume with 14,888 units, followed by Supreme (11,987), Veggie (11,649), and Chicken (11,050) — a fairly balanced spread.

```sql
-- Join the necessary tables to find the total quantity of each pizza category ordered.

Select pizza_types.category, sum(orders_details.QUANTITY) as quantity
from orders_details join pizzas on orders_details.PIZZA_ID = pizzas.pizza_id
join pizza_types on pizzas.pizza_type_id = pizza_types.pizza_type_id
group by pizza_types.category
order by quantity desc;
```



INTERMEDIATE · Q7

Determine the distribution of orders by hour of the day.

```sql
-- Determine the distribution of orders by hour of the day.
SELECT
    HOUR(orders.order_time) AS dayhour,
    SUM(orders_details.QUANTITY) AS quantity
FROM
    orders_details
        JOIN
    orders ON orders_details.ORDER_ID = orders.ORDER_ID
GROUP BY dayhour
ORDER BY quantity DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dayhour	quantity
12	6776
13	6413
18	5417
17	5211
19	4406
16	4239
14	3613
20	3534
15	3216
11	2728
21	2545
22	1386
23	68
10	18
9	4

KEY INSIGHT

Orders peak sharply at 12 PM (6,776 pizzas) and again at 1 PM and 6 PM, pointing to clear lunch and dinner rushes worth staffing around.

INTERMEDIATE · Q8

Join relevant tables to find the category-wise distribution of pizzas.

KEY INSIGHT

The menu itself is well balanced across categories: Veggie and Supreme offer 9 pizza types each, Classic 8, and Chicken 6 — variety is fairly even.

```
1   -- Join relevant tables to find the category-wise distribution of pizzas.
2 • Select category, count(name)
3   from pizza_types
4   group by category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	count(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

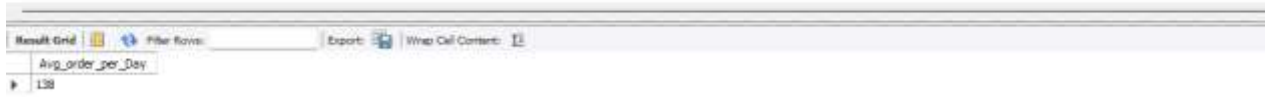
INTERMEDIATE · Q9

Group the orders by date and calculate the average number of pizzas ordered per day.

KEY INSIGHT

On average, about 138 pizzas are sold per day, calculated using a subquery that first totals daily quantity before averaging across all order dates.

```sql
-- Group the orders by date and calculate the average number of pizzas ordered per day.
SELECT
    ROUND(AVG(quan)) AS Avg_order_per_Day
FROM
    (SELECT
        orders.ORDER_DATE, SUM(orders_details.QUANTITY) AS quan
    FROM
        orders_details
    JOIN orders ON orders.ORDER_ID = orders_details.ORDER_ID
    GROUP BY orders.ORDER_DATE) AS order_per_day;
```



INTERMEDIATE · Q10

Determine the top 3 most ordered pizza types based on revenue.

KEY INSIGHT

By revenue, Thai Chicken ($43,434), Barbecue Chicken ($42,768), and California Chicken ($41,410) top the list — none of which are the top sellers by quantity, showing higher price points drive more revenue per order.

```sql
-- Determine the top 3 most ordered pizza types based on revenue.
SELECT
    pizza_types.name,
    ROUND(SUM(orders_details.QUANTITY * pizzas.price),
            0) AS revenue
FROM
    pizzas
        JOIN
    orders_details ON pizzas.pizza_id = orders_details.PIZZA_ID
        JOIN
    pizza_types ON pizza_types.pizza_type_id = pizzas.pizza_type_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

name	revenue
The Thai Chicken Pizza	43434
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41410

03

Advanced Queries

Window functions and CTEs to uncover revenue share, trends, and category leaders

ADVANCED · Q11

Calculate the percentage contribution of each pizza type to total revenue.

KEY INSIGHT

Using a CTE to hold total revenue, Classic contributes 26.91% of all sales, followed closely by Supreme (25.46%), Chicken (23.96%), and Veggie (23.68%) — no single category dominates.

```sql
-- Calculate the percentage contribution of each pizza type to total revenue.
With total as ( SELECT
    SUM(orders_details.QUANTITY * pizzas.price) AS Total_Revenue
FROM orders_details JOIN
    pizzas ON orders_details.PIZZA_ID = pizzas.pizza_id
    )
SELECT
    pizza_types.category,
    Round((SUM(orders_details.QUANTITY * pizzas.price) / total.Total_Revenue ) * 100,2) as revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    orders_details ON orders_details.PIZZA_ID = pizzas.pizza_id
        CROSS JOIN
        total
GROUP BY pizza_types.category, total.Total_Revenue
ORDER BY revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	revenue
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

ADVANCED · Q12

Analyze the cumulative revenue generated over time.

KEY INSIGHT

A running total built with SUM() OVER (ORDER BY order_date) shows steady, consistent daily growth — revenue crosses $52,700 within the first 23 days of the year with no major dips.

```sql
-- Analyze the cumulative revenue generated over time.
Select ORDER_DATE, Round(sum(revenue) Over(order by order_date),2) as cum_revenue
From(
Select orders.ORDER_DATE,
sum(orders_details.QUANTITY * pizzas.price) as revenue
from pizzas join orders_details on pizzas.pizza_id = orders_details.PIZZA_ID
join orders on orders.ORDER_ID = orders_details.ORDER_ID
group by orders.ORDER_DATE ) as daily_revenue
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ORDER_DATE	cum_revenue
2015-01-01	2713.85
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05
2015-01-09	21526.4
2015-01-10	23990.35
2015-01-11	25862.65
2015-01-12	27781.7
2015-01-13	29831.3
2015-01-14	32358.7
2015-01-15	34343.5
2015-01-16	36937.65
2015-01-17	39001.75
2015-01-18	40978.6
2015-01-19	43365.75
2015-01-20	45763.65
2015-01-21	47804.2
2015-01-22	50300.9
2015-01-23	52724.6

ADVANCED · Q13

Determine the top 3 most ordered pizza types based on revenue for each pizza category.

KEY INSIGHT

A CTE plus DENSE_RANK() OVER (PARTITION BY category ORDER BY revenue DESC) ranks pizzas within their own category — surfacing each category's top 3 earners side by side, e.g. Thai Chicken leads Chicken, Classic Deluxe leads Classic.

```sql
-- Determine the top 3 most ordered pizza types based on revenue for each pizza category.
WITH PizzaRevenue AS (
    SELECT
        pizza_types.category,
        pizza_types.name,
        Round(SUM(orders_details.QUANTITY * pizzas.price)) AS Revenue
    FROM pizzas
    JOIN orders_details
        ON pizzas.pizza_id = orders_details.PIZZA_ID
    JOIN pizza_types
        ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    GROUP BY pizza_types.category, pizza_types.name
),
RankedPizza AS (
    SELECT
        category,
        name,
        Revenue,
        DENSE_RANK() OVER (PARTITION BY category ORDER BY Revenue DESC) AS drnk
    FROM PizzaRevenue
)
SELECT
    category,
    name,
    Revenue,
    drnk
FROM RankedPizza
WHERE drnk <= 3;
```

category	name	Revenue	drnk
Chicken	The Thai Chicken Pizza	43434	1
Chicken	The Barbecue Chicken Pizza	42768	2
Chicken	The California Chicken Pizza	41410	3
Classic	The Classic Deluxe Pizza	38180	1
Classic	The Hawaiian Pizza	32273	2
Classic	The Pepperoni Pizza	30162	3
Supreme	The Spicy Italian Pizza	34831	1
Supreme	The Italian Supreme Pizza	33477	2
Supreme	The Sicilian Pizza	30940	3
Veggie	The Four Cheese Pizza	32266	1
Veggie	The Mexicana Pizza	26781	2
Veggie	The Five Cheese Pizza	26066	3

THANK YOU

Pizza Sales Analysis — MySQL Query Portfolio

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